

See [Section F.1, “Certification Declaration CMS test vector”](#) for a complete example of generating a Certification Declaration.

6.3.2. Firmware Information

Firmware Information is an optional component of the [Device Attestation Information](#) (see [Section 6.2.3, “Device Attestation Procedure”](#)).

Firmware Information MAY contain one or more Firmware Digests that correspond to the components in the firmware that have been measured and recorded during the boot process (e.g., bootloader, kernel, root filesystem, etc), and MAY contain other metadata. A Firmware Digest SHALL either represent a hash of the corresponding firmware layer or a hash of the signed manifest that was used to validate the corresponding firmware layer during secure boot. An implementation MAY choose to hash the measurements of all components into a single hash and include only that hash in Firmware Information.

A device MAY report Firmware Information containing its firmware digests only if it implements a secure subsystem that protects the device attestation private key and is able to securely collect and report firmware digests as shown in [Figure 53, “Illustration of the measured boot process”](#). This process is known as "measured boot".

Ideally, the measured boot process SHOULD be rooted in silicon such as a boot ROM, similar to the secure boot process found in many systems-on-chip (SoCs). Since many SoCs and microcontrollers are unable to perform measured boot in hardware, the process SHOULD start at the earliest firmware component possible (for example, at the bootloader shown in the figure below). In this case, this firmware component is not measured and in fact, it is the root for measurement. Therefore, it SHALL be resistant to attacks compromising subsequent firmware components (e.g., the ROM must verify its authenticity (secure boot) or it may be placed in a locked partition at the factory that cannot be updated by software in the field).

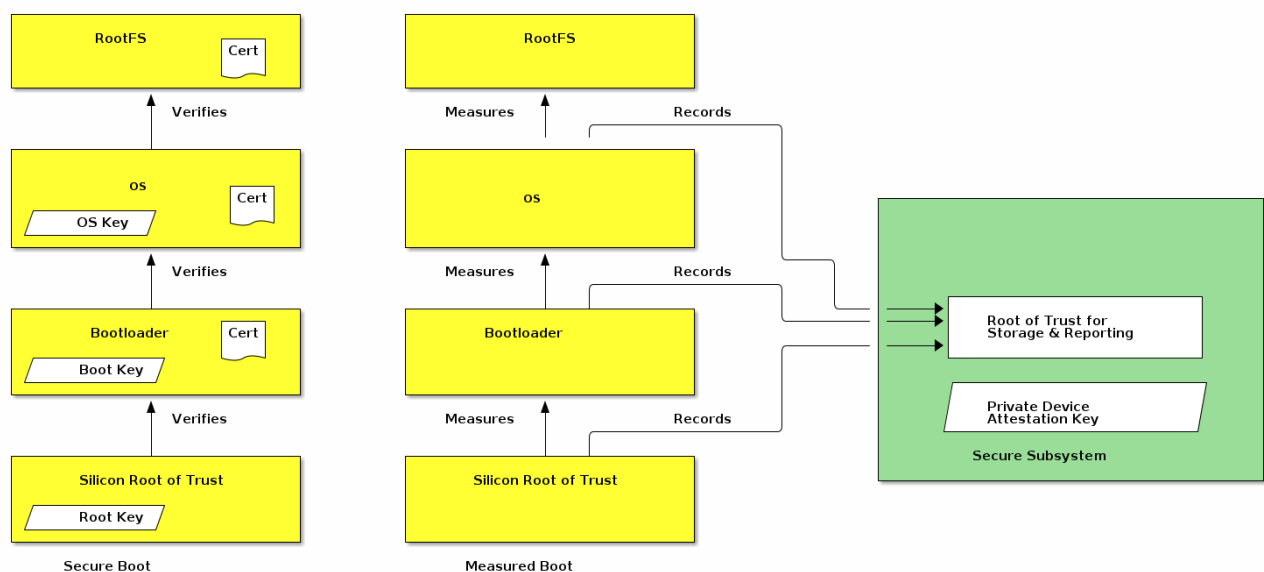


Figure 53. Illustration of the measured boot process

The device secure subsystem SHALL use the device attestation private key to sign [attestation-elements](#) and [NOCSR-elements](#). The device secure subsystem SHALL fill the [attestation-elements](#) fields